

SPI Controller — Verification Plan

Reset & Enable Control

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
PRESETn [R2]	Registers and FIFOs reset to default values after reset assertion	Assert/deassert reset during normal and randomized operation	Reset values, FIFO reset, BUSY clear, SS_n idle	Verify register reset values, FIFO empty, BUSY=0, SS_n=1111
ctrl_en [R3]	SPI disabled when CTRL.EN=0	Disable SPI then attempt TX/FIFO operations	Disabled-state behavior, FIFO status, SS_n idle	Verify TX/RX empty, BUSY=0, SS_n high, no transfer activity

SCLK Generation & Clock Divider

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
SCLK Divider Operation [R8] [R24]	SCLK frequency scales with CLK_DIV value and remains stable during transfers.	Apply corner and randomized divider values during SPI transfers.	Divider values, SCLK frequency scaling, active transfers.	Verify SCLK frequency changes correctly and remains stable.
SCLK Idle Polarity [R4]	SCLK idle polarity matches configured SPI mode.	Sweep SPI modes with different transfer settings.	SPI modes, idle SCLK states, idle polarity behavior.	Verify modes 0/1 idle LOW and modes 2/3 idle HIGH.

Data Transfer — Width, Bit-Order & BUSY

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
Transfer width control [R7]	SPI transfer length follows configured width setting.	Apply directed and randomized transfers with different widths	Width configurations, BUSY behavior, illegal width cases	Verify correct transfer length, BUSY timing, and illegal width fallback
Bit order control [R6]	MOSI transmission follows selected LSB/MSB order.	Sweep SPI modes, widths, and bit-order settings.	LSB/MSB configurations, first-bit behavior, loopback data	Verify transmitted first bit and correct TX/RX ordering

MOSI [R5]	MOSI changes only on launch edges and remains stable on sampling edges.	Generate transfers across SPI modes with randomized TX data	MOSI behavior, idle state, transfer configurations	Verify MOSI stability, edge alignment, and no glitches
Spi mode operation cpol / cpha [R4][R5]	All four SPI modes (0–3) must be correctly implemented. CPOL=mode[1], CPHA=mode[0]. Mode 0: sample ↑, launch ↓. Mode 1: sample ↓, launch ↑. Mode 2: sample ↓, launch ↑. Mode 3: sample ↑, launch ↓.	Sweep all SPI modes with different widths and bit orders	SPI modes, width/order combinations, loopback scenarios	Verify correct sampling/launch edges and SCLK polarity for all modes

Loopback Mode

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
Loopback mode [R19]	Transmitted MOSI data is internally looped back to RX path when loopback is enabled	Run loopback transfers across SPI modes with randomized TX data	Loopback operation, SPI modes, widths, bit orders	Verify RX data matches transmitted TX data and external MISO is ignored

TX & RX FIFO – Depth, Status & Overflow

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
Tx_FIFO status [R11]	TX FIFO depth is 8 entries with correct TX_FULL and TX_EMPTY indication	Generate directed FIFO fill, drain, and overflow sequences	Cover TX FIFO fill levels, full, empty, and overflow conditions.	Verify TX_FULL, TX_EMPTY, and TX overflow behavior
Rx_fifo_status [R12]	RX FIFO depth is 8 entries with correct RX_FULL and RX_EMPTY indication	- Generate RX fill, drain, and overflow scenarios	Cover RX FIFO occupancy, full, empty, and overflow conditions.	Verify RX_FULL, RX_EMPTY, and RX overflow behavior
tx_push_dropped [R13]	TX overflow writes are discarded without corrupting FIFO contents	- Directed FIFO overflow and recovery sequences.	Cover TX FIFO overflow conditions and TX_OVF interrupt states.	Verify overflowed writes are ignored and FIFO ordering is preserved
RX FIFO full condition RX FIFO write operation [R14]	RX overflow sets RX_OVF while preserving existing FIFO data	- Generate RX overflow without RX reads	Cover RX FIFO overflow conditions and RX_OVF interrupt behavior.	Verify correct FIFO data ordering after overflow and proper clearing of INT_STAT[3] using W1C

rx_empty_read [R15]	Reading RX_DATA while RX FIFO is empty returns default value without affecting FIFO state.	- Directed empty RX read scenarios after FIFO drain.	Cover RX empty-read behavior.	Verify RX_DATA returns 0x00000000 and FIFO status/interrupts remain unchanged.
------------------------	--	--	-------------------------------	--

Interrupt System – INT_EN, INT_STAT, IRQ, W1C

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
Interrupt Generation IRQ [R16]	IRQ = (INT_STAT & INT_EN) at all times. Bits: [0]=TX_EMPTY, [1]=RX_FULL, [2]=TX_OVF, [3]=RX_OVF, [4]=TRANSFER_DONE.	- Directed interrupt enable/disable and event generation sequences with IRQ verification.	Cover interrupt set/clear/mask behavior and IRQ response for all interrupt sources..	Check IRQ generation, masking, and unmasking behavior against interrupt status and enable bits.
W1C Register Behavior [R17]	INT_STAT W1C behavior: Verify bits clear on write-1 and remain unchanged on write-0.	- Directed interrupt clear and W1C status clearing sequences.	Cover INT_STAT W1C clear operations and cleared bit combinations.	Verify only written '1' bits in INT_STAT are cleared, while written '0' bits remain unchanged.
W1C Race Handling [R18]	Verify HW priority during simultaneous W1C clear and interrupt event assertion for TX_OVF, RX_OVF, RX_FULL, TX_EMPTY, and TRANSFER_DONE.	- Directed simultaneous interrupt clear and event generation scenarios.	Cover interrupt set/clear race conditions.	Verify W1C race handling by ensuring interrupt bits remain asserted during simultaneous clear and hardware event conditions. .

Slave-Select (SS_n) & APB Protocol

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
SS_CTRL / SS_n Generation [R20]	SS_n[i] = ~(ss_en[i] & ss_val[i]) during active transfer. SS_n remains LOW throughout enabled transactions.	Apply directed SS_CTRL read/write and transfer sequences	Cover SS_CTRL accesses and SS_n assertion/deassertion behavior	Verify correct SS_n assertion/deassertion during transfers
APB Signals [R22]	APB protocol follows correct SETUP/ACCESS timing with zero wait states	Generate APB read/write transaction sequences	APB accesses and protocol behavior	Verify APB timing, stable signals, and valid read/write operations
Reserved Address Access [R23]	Reserved APB offsets return zero on reads and ignore writes without side effects.	Directed reserved-address read/write accesses	Cover reserved address access behavior	Verify invalid reads return zero and reserved writes do not affect valid registers

Inter-Transfer Delay & Configuration Latching

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
INTER_TRANSFER_DELAY [R21]	Delay cycles inserted between consecutive transfers based on DELAY value	Apply directed and randomized delay configurations during transfers	Delay values, GAP transitions, TX empty conditions	Verify correct gap insertion and no GAP when delay is zero or TX FIFO is empty
CFG_LATCH [R25]	Transfer settings are latched at transfer start and remain unchanged during active transfer	Change CTRL settings during and between transfers	Configuration latch behavior during BUSY state	Verify active transfer settings remain stable while BUSY is asserted

APB Register Map – Read/Write Access & Reset Values

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
CTRL 0x00	R/W. [0] EN · [1] MSTR · [3:2] MODE · [4] LSB_FIRST · [5] LOOPBACK · [7:6] WIDTH. Reset: 0x00000000.	- Directed CTRL register access sequences across mode, enable, and width configurations. - Randomized CTRL register configurations and accesses.	Cover CTRL register read/write combinations and reset behavior.	Verify correct register read/write behavior and reset default value.
STATUS 0x04	Read-only. [0] BUSY · [1] TX_FULL · [2] TX_EMPTY · [3] RX_FULL · [4] RX_EMPTY · [5] TX_OVF · [6] RX_OVF. Reset: 0x00000014 (TX_EMPTY=1, RX_EMPTY=1).	-STATUS polling during SPI activity. - Directed status-flag checking sequences.	Cover STATUS register reset value (0x14) and status bit updates.	Verify STATUS reset value and ensure write attempts do not modify the register contents.
TX_DATA 0x08	Write-only (WO). Reads return 0x00000000. Written value pushed to TX FIFO if not full.	- Directed invalid write/read sequences for RO registers. - Randomized register write sequences.	Cover WO register access and TX FIFO push operations.	Verify WO register reads return 0, valid writes push into TX FIFO, and writes are discarded when FIFO is full.
RX_DATA 0x0C	Read-only (RO). Pops one entry from RX FIFO. Returns 0x00000000 when empty. Writes ignored.	-Directed RX_DATA read sequences after transfers. - Directed empty RX read scenarios.	Cover RX FIFO pop operations and empty read behavior.	Verify FIFO read ordering, empty reads return 0, empty reads do not assert RX_OVF, and write attempts are ignored.
CLK_DIV 0x10	R/W. [15:0] divider value. Upper 16 bits reserved (read as 0). Reset: 0x00000000.	-Directed register writes, clock-divider corner case values, and constrained-random register accesses in the range [1–32] for functional and robustness coverage	Cover CLK_DIV register writes accesses and divider value ranges.	Verify correct CLK_DIV read/write behavior, reserved bits cleared, and proper SCLK generation.

SS_CTRL 0x14	R/W. [3:0] ss_en · [7:4] ss_val. Drives SS_n combinationaly. Reset: 0x00000000.	All tests use proper chip-select control, with directed register write access (e.g., 0xA5) for basic APB functionality validation.	Cover SS_CTRL read/write accesses and SS_n generation behavior.	Verify SS_CTRL read/write behavior and correct SS_n generation.
INT_EN 0x18	R/W. [4:0] enable mask for 5 interrupt sources. Reset: 0x00000000. Upper bits reserved (read as 0).	- Directed interrupt enable configurations with selected bit patterns. - Constrained-random register accesses W1C and robustness verification.	Cover INT_EN read/write accesses and interrupt enable/disable states.	Verify correct register read-back, reserved bits cleared to 0, and IRQ masking behavior.
INT_STAT 0x1C	Read / W1C. [4:0] sticky interrupt flags. Reset: 0x00000000. Writing 1 clears the bit; writing 0 no effect. HW events set bits regardless of mask.	- Directed interrupt generation, W1C clearing, and interrupt race-condition scenarios. - Randomized W1C stress and clear-validation sequences.	Cover interrupt event generation, interrupt bits, and IRQ assertion states.	Verify W1C behavior for R17_W1C and R18_W1C_RACE_IF2–5 using check_int_bit() after each W1C write. Verify write-0 operation causes no register change.
DELAY 0x20	R/W. [7:0] inter-transfer delay count. Reset: 0x00000000. Upper 24 bits reserved (read as 0).	-Directed register and transfer stimulus with predefined data patterns and delay configurations. - Constrained-random delay generation in the range [0:255].	Cover DELAY register read/write accesses, delay value ranges (zero, non-zero, max), and GAP state entry/exit conditions.	Verify read-back data [7:0] correctness; upper bits are zero. Validate gap timing = DELAY × (DIV + 1) PCLK cycles.

Transfer Flow – BUSY & State Machine

Label	Design Requirement	Stimulus Generation	Functional Coverage	Functionality Check
busy_and_state_flow [R7]	BUSY remains asserted during active transfer states (S_SHIFT, S_FINISH, S_GAP) and clears after transfer completion	Run directed and randomized transfers with different widths and divider values	BUSY behavior during active and idle states	Verify correct BUSY assertion/deassertion and transfer completion behavior
transfer_done_pulse [R7]	transfer_done_pulse is asserted for exactly one cycle at transfer completion and triggers RX FIFO push and TRANSFER_DONE interrupt	Generate transfers with different widths and bit orders	Transfer-done pulse and interrupt behavior	Verify done_pulse race condition (R18_W1C). Confirm interrupt bit [4] sets per transfer. Ensure RX FIFO increments once per completed transfer and TRANSFER_DONE interrupt asserts correctly.